

Non-linear dynamics and chaos

1 Lecture 1 - Introduction and overview

Chaos

An aperiodic and seemingly unpredictable phenomenon that occurs in deterministic systems. Sensitive dependence on initial conditions.

Poincare sucked at drawing. So, it was hard to decipher his work. Check out Lorenz System, and his paper. Chaos theory gained momentum in 1975, work on iterated maps $x_{n+1} = f(x_n)$. Another kind of map is logistic map,

$$x_{n+1} = rx_n(1 - x_n)$$

where, r is the control parameter. Changing r changes the long-term behavior: stable equilibrium, oscillations, or chaos.

Mandelbrot fractals

The Mandelbrot set is defined in the complex plane by iterating the function,

$$z_{n+1} = z_n^2 + c$$

where,

- $z_0 = 0$
- c is a complex parameter
- A point c belongs to the Mandelbrot set if the sequence z_n does not diverge to infinity as $n \rightarrow \infty$

When plotted, the Mandelbrot set is a fractal – a shape with

- infinite detail,
- self-similarity,
- complex boundaries.

It is famous for its “cauliflower” main shape with infinite smaller copies of itself embedded inside. Zooming in reveals endlessly intricate structures. Applications include modeling patterns in nature such as coastlines, clouds, turbulence, and even finance.

Mandelbrot explored many fractals beyond the Mandelbrot set, such as:

- Julia sets: (closely related to the Mandelbrot set, defined for fixed values of c),
- Self-similar fractals like the Koch snowflake, Cantor set, and Sierpiński triangle,
- Fractals in nature – trees, rivers, mountains, and markets.

Check out Arthur T. Winfree’s work on non-linear oscillators is central to modern biological rhythm research and synchronization theory.

Ruen and Takens explored whether Chaos and Turbulence are similar.

In 1978 – Feigenbaum discovered that as the control parameter increases, the system undergoes period-doubling bifurcations on the route to chaos – universal route to chaos. Explained it through connections to phase transitions and statistical physics.

1.1 Mathematical Definition of the Renormalization Group

RG as a Transformation

Suppose a system is described by a Hamiltonian (or action, or map)

$$H(\{g_i\}),$$

where $\{g_i\}$ are parameters (coupling constants, control parameters, etc.).

The *renormalization group transformation* is an operator $\mathcal{R}_b$ (with scale factor $b > 1$) that acts as:

$$H' = \mathcal{R}_b(H),$$

such that H' describes the same physics as H , but with short-distance (microscopic) degrees of freedom integrated out and lengths rescaled by b .

RG Flow in Coupling Space After applying $\mathcal{R}_b$, the couplings change:

$$\{g'_i\} = \mathcal{R}_b(\{g_i\}).$$

If we iterate this process, we get a *flow* in parameter space. Taking the continuous limit, the flow is governed by *beta functions*:

$$\frac{dg_i}{d \ln b} = \beta_i(\{g\}),$$

where β_i encodes how the coupling g_i changes with scale.

Fixed Points A point $\{g_i^*\}$ is a *fixed point* of the RG if:

$$\beta_i(\{g^*\}) = 0 \quad \text{for all } i.$$

- Stable fixed points $\rightarrow$ attract nearby flows (universal behavior at large scales).
- Unstable fixed points $\rightarrow$ describe phase transitions (critical points).

Example: Logistic Map & Feigenbaum RG

For the logistic map

$$f_r(x) = rx(1 - x),$$

the RG transformation looks like:

$$\mathcal{R}[f](x) = \alpha f(f(x/\alpha)),$$

where α is a rescaling constant.

The fixed point of this functional equation explains *Feigenbaum's universality constant* ($\delta \approx 4.669$).

5. Compact Definition The *renormalization group* is the semi-group of transformations $\mathcal{R}_b$ acting on Hamiltonians (or dynamical maps), defined by coarse-graining and rescaling, such that:

$$\mathcal{R}_b : \{g_i\} \mapsto \{g'_i(b)\}, \quad \frac{dg_i}{d \ln b} = \beta_i(\{g\}),$$

with *fixed points* $\{g^*\}$ determining universal large-scale behavior.

1.2 Logical structure of dynamics

Differential equation of form,

$$\underline{\dot{x}} = \underline{f}(\underline{x})$$